Multiobjective Multitasking Evolutionary Optimization with Adaptive Knowledge Transfer

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***Abstract*—Evolutionary algorithms have become the mainstream method for solving multiobjective optimization problems (MOPs) due to their powerful search performance. This paper proposes a multiobjective multitasking evolutionary algorithm with self-adaptive knowledge transfer, called SAMO-MFEA. Specifically, it can capture inter-task synergy based on the feedback of the evolutionary population to realize adaptive regulation of knowledge transfer.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

In recent years, inspired by multi-task learning, the concept of evolutionary multitasking optimization (EMTO) has been proposed in the field of evolutionary computation [1]. Different from traditional evolutionary algorithms, EMTO aim to solve several different optimization problems simultaneously, exploit the similarity between tasks, and improve the overall optimization efficiency by sharing and reusing what the optimizer has already learned. Recently, EMTO has been promoted and improved by lots of studies. The practicality of EMTO has also been demonstrated in a range of real-world applications such as robot controller design and path planning [2]. However, the applicability and success of current multitasking solvers depend heavily on task relevance. For those optimization problems that have little in common, blindly transferring knowledge between them may lead to primarily negative inter-task interactions [3].

II. Framework of SAMO-MFEA

Multifactor optimization (MFO) is a popular EMTO paradigm in which groups of different offspring assigned to different tasks are allowed to transfer knowledge [3]. In MFO, all individuals are encoded into a uniform representation space. In order to compare the performance of individuals, several attributes associated with individuals are defined as follows. *Definition 1 (Factorial Rank)*: The factorial rank r_i^k of p_i for task T_k is the index of p_i in the list of population members sorted in decreasing order of preference with respect to T_k. This paper evaluates the factor rank of each individual based on non-dominated front (NF) and crowded distance (CD). *Definition 2 (Skill Factor)*: The skill factor is defined by the index of the task which an individual is assigned to. Skill factor of p_i is given by $\tau_i = \arg\min_k r_k^i$, where $k \in \{1, 2, \ldots, K\}$. *Definition 3 (Scalar Fitness)*: The scalar fitness φ_i (or note as $\varphi(p_i)$) of p_i is the inverse of $r_i^{\tau_i}$ given by $\varphi_i = 1/r_i^{\tau_i}$.

The strength of knowledge transfer between different tasks is regulated by a random mating parameter, denoted as *RMP*. Different from traditional EMTO methods, the proposed SAMO-MFEA expands the *RMP* into a matrix whose elements are able to measure the correlation between tasks.

To facilitate positive knowledge transfer between tasks and avoid the threat of negative transfer, SAMO-MFEA is proposed. In the proposed method, the *RMP* is no longer a scalar parameter but takes the form of an asymmetric $K \times K$ matrix of pairwise values (given K optimization tasks).The elements of the t_1 th row and t_2 th column of the matrix are denoted as rmp_{t_1,t_2}, where $[t_1, t_2] \in \{1, 2, ..., K\}$. It represents the synergistic effect of task t_1 on task t_2.

In SAMO-MFEA, the offspring generated by the knowledge transfer-based generation approach and those generated by the direct mutation-based approach are recorded in S and S', respectively. Our goal is to assess the relative advantages of the two methods in real time and then dynamically adjust *RMP* based on the assessment results. Firstly, for an offspring in the set $S \cup S'$, the level of improvement (LoI) is computed when it outperforms its parent. It is worth noting that since only comparisons of performance between individuals belonging to the same task are significant, only the parent with the same skill factor as the offspring is selected for the computation of LoI. The computation of LoI can be represented as:

TABLE I
PERFORMANCE COMPARISON (IGD MEAN AND BRACKETED STANDARD DEVIATION) OF ASMO-MFEA AND THREE ADVANCED ALGORITHMS{NSGA-II, MO-MFEA, MO-MFEA-II} ON CEC17-MTMO.

Task No	NSGA-II	MO-MFEA	MO-MFEA-II	SAMO-MFEA
CEC17-MTMO1-CI-HS-T1	4.5587e-02 (6.78e-03) -	1.4147e-02 (3.88e-03) -	1.0504e-02 (2.06e-03) -	**4.7416e-03 (1.09e-04)**
CEC17-MTMO1-CI-HS-T2	1.1363e-01 (1.34e-02) -	8.7770e-02 (1.67e-02) -	7.2995e-02 (1.13e-02) -	**9.8911e-03 (2.32e-03)**
CEC17-MTMO2-CI-MS-T1	1.3766e-02 (6.23e-03) -	1.0011e-02 (9.05e-03) -	9.3210e-03 (6.26e-03) -	**4.4828e-03 (1.18e-04)**
CEC17-MTMO2-CI-MS-T2	8.0701e-01 (7.10e-01) -	1.6039e-02 (1.08e-02) -	1.4661e-02 (7.36e-03) -	**8.1183e-03 (6.88e-03)**
CEC17-MTMO3-CI-LS-T1	8.4012e+00 (2.82e+00) -	9.1931e-03 (1.01e-03) -	**6.1288e-03 (5.28e-04) =**	6.2336e-03 (1.31e-03)
CEC17-MTMO3-CI-LS-T2	5.2755e-03 (2.11e-04) -	5.5399e-03 (2.38e-04) -	4.6452e-03 (1.36e-04) -	**4.2972e-03 (9.65e-05)**
CEC17-MTMO4-PI-HS-T1	3.3690e-02 (6.50e-03) -	1.3956e-02 (4.44e-03) -	1.6171e-02 (6.40e-03) -	**4.3373e-03 (1.51e-04)**
CEC17-MTMO4-PI-HS-T2	1.2444e+00 (5.94e-01) =	**8.7983e-01 (4.22e-01) =**	1.2325e+00 (1.02e+00) =	7.3218e+00 (8.52e+00)
CEC17-MTMO5-PI-MS-T1	1.4268e-01 (6.16e-02) =	**7.9118e-02 (3.16e-02) +**	1.4701e-01 (4.60e-02) =	1.3105e-01 (4.85e-02)
CEC17-MTMO5-PI-MS-T2	4.6418e+02 (6.24e+01) -	4.0076e+02 (1.24e+02) -	3.7493e+02 (1.36e+02) =	**2.5531e+02 (7.86e+01)**
CEC17-MTMO6-PI-LS-T1	7.4524e-03 (3.99e-03) -	8.3040e-03 (2.48e-03) -	6.1381e-03 (1.95e-03) -	**4.7327e-03 (1.73e-04)**
CEC17-MTMO6-PI-LS-T2	2.0032e+01 (1.14e-02) -	**3.7321e+00 (7.43e+00) +**	2.0027e+01 (2.06e-02) -	2.0002e+01 (4.36e-02)
CEC17-MTMO7-NI-HS-T1	1.4281e+03 (2.64e+03) -	4.8668e+01 (8.16e-01) =	**4.5513e+01 (4.89e+01) =**	5.5234e+01 (1.54e+01)
CEC17-MTMO7-NI-HS-T2	2.2256e-02 (4.61e-03) -	1.3938e-02 (3.65e-03) -	1.2048e-02 (3.49e-03) -	**4.2710e-03 (1.01e-04)**
CEC17-MTMO8-NI-MS-T1	4.9050e+01 (3.42e+01) -	3.3464e+01 (3.29e+01) =	5.2863e+01 (3.73e+01) -	**1.2980e+01 (2.81e+00)**
CEC17-MTMO8-NI-MS-T2	2.2030e+00 (1.57e+00) -	1.8484e+00 (2.67e+00) -	1.6096e+00 (9.51e-01) -	**4.7321e-01 (8.28e-01)**
CEC17-MTMO9-NI-LS-T1	6.9454e-02 (3.00e-03) -	7.4172e-02 (3.64e-03) -	6.9320e-02 (3.71e-03) =	**6.6979e-02 (3.06e-03)**
CEC17-MTMO9-NI-LS-T2	2.0295e+01 (2.59e-04) +	2.0301e+01 (6.62e-03) =	**2.0295e+01 (2.45e-04) +**	2.0298e+01 (1.05e-02)
+ / - / =	1 / 15 / 2	2 / 12 / 4	1 / 11 / 6	Base

$$LoI_i = |\varphi(c_i) - \varphi(p_i)| \;/\; \varphi(p_i), \quad (1)$$

where $\varphi(c_i)$ represents the scalar fitness of c_i. When all rmp are obtained, the effectiveness of knowledge transfer ($EoKT$) between each task is calculated and can be written as:

$$EoKT_{t_1,t_2} = \frac{\sum_{i \in S_{t_1,t_2}} LoI_i / |S_{t_1,t_2}|}{\sum_{i \in S_{t_1,t_2}} LoI_i / |S_{t_1,t_2}| + \sum_{j \in S'} LoI_j / |S'|}, \quad (2)$$

where $S_{t_1,t_2} \subset S$, which stores the offspring individuals with skill factor τ_{t_2} generated by two parents with skill factors τ_{t_1} and τ_{t_2}. The individuals in S_{t_1,t_2} can represent the knowledge transfer from task t_1 to t_2. The larger the value of $EoKT$, the higher the probability of obtaining an improvement using the knowledge transfer approach to generate offspring relative to the direct mutation approach. Therefore, $EoKT$ is employed to perform the update of *RMP*, which can be written as:

$$RMP_{t_1,t_2} = rmp_{t_1,t_2} + w \times (EoKT_{t_1,t_2} - rmp_{t_1,t_2}), \quad (3)$$

where w represents the rate at which *RMP* is regulated.

III. EXPERIMENTAL STUDY

To conduct the experimental study, the benchmark suite CEC17-MTMO with nine sets of two-task multiobjective test problems is employed. For more information on the CEC17-MTMO test refer to [4].

The proposed method is compared with two multiobjective EMTO algorithms, MO-MFEA [5] and MO-MFEA-II [6], and the classical multiobjective single-task algorithm, NSGAII [7]. MO-MFEA is the baseline of the multiobjective EMTO algorithm, which uses the well-known NSGA-II as an evolutionary engine to solve MOPs. MO-MFEA-II is an improved version of MO-MFEA.

Table I summarizes the optimal values and variances of Inverse Generational Distance (IGD) metrics obtained by all algorithms. It can be observed that the two MTO algorithms (i.e., MO-MFEA, ,MO-MFEA-II) outperform NSGA-II. These results confirm that inter-task knowledge transfer can accelerate the convergence of these problems. The results of nine sets of test problems show that the proposed method outperforms MO-MFEA and MO-MFEA-II in solving most of the problems. In addition, the proposed method achieves optimal results on 12 of the 18 multiobjective optimization tasks.

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